

TPU TYRE DESIGN

Presents for UGV-UAV's System for payload

Why TPU and not normal rubber tyres?

- Cheaper than rubber tyres
- No puncture risks and overall less weight (this improves motor life as well)
- Terrain specific customization
- Damping reduces electronic shock
- For rigid chassis, TPU wheels are good for shock absorption.

Why using treads? : Basically for traction and durability.

—> These displace debris, increase surface contact on inclines, and improve wet/dry skid resistance. For climbing, deeper treads (5-10mm) allow the tire to conform to surfaces, boosting friction.

TREAD DESIGN:

Chevron/ V-lugs —> best for ramps + uneven terrains

Unidirectional treads

Abrasion resistant so that tyre ka material scrape off na ho



Recommended Dimensions assuming the wheel diameter is 30cm

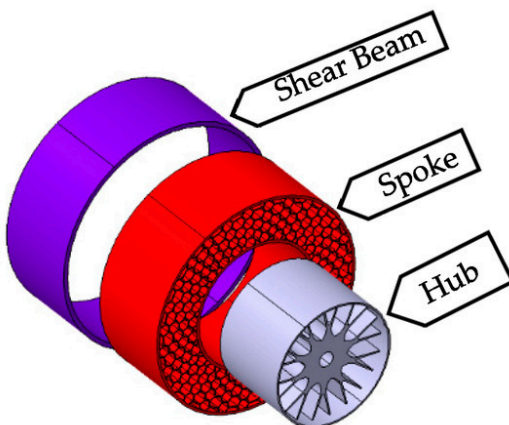
Angle: 30-45 deg

Lug height: **8–12 mm**

Lug thickness: **≥6 mm**

Lug spacing: **1.2–1.5 × lug width**

Rounded edges (fillet radius **≥2 mm**)

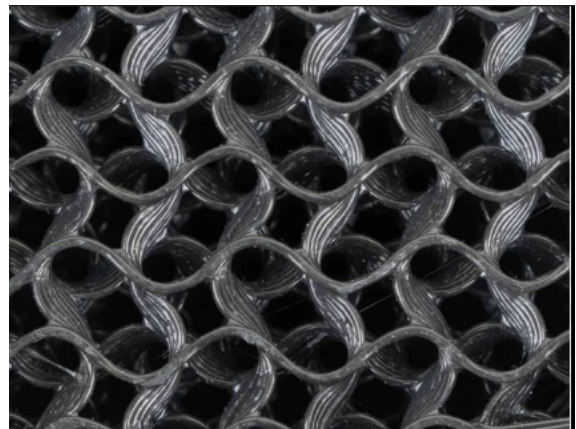
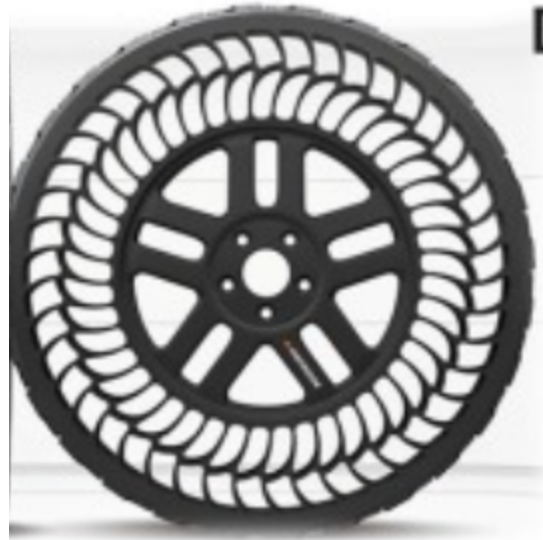


TPU shore should be around 85-90A
(balance between soft and hard) Good
for a 50kg UGV
(TPU has hardness chart depending
upon its use)
This part needs to be researched more.

HUB DESIGN:

Rigid hub with proper locking method.
Material - Aluminium or nylon CF for
robust attachment of motor and so that it
can handle the torque

SPOKES DESIGN:



CURVED RADIAL SPOKES

GYROID SPOKES

These infill patterns will help in shock
absorption

Radial pattern is considered best for
heavy UGVs

Void ratio should be 35-50%

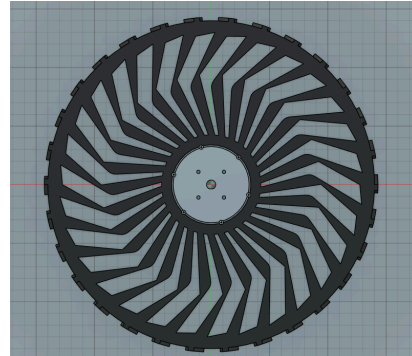
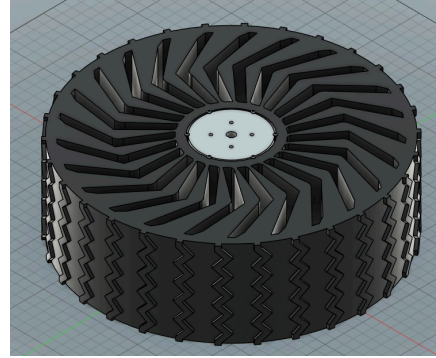
What are radial spokes? – spokes
that run **directly from hub to tread**,
roughly normal to the hub. The spoke
geometry determines **how forces flow**
inside the wheel.

**Why curved radial spokes are
best?**

1. Each spoke compresses independently
2. Load spreads over multiple spokes
3. Progressive stiffness (soft at first, stiffer later)
4. Better fatigue life (good for repeated tests in rough terrains)

This reduces:

- Peak G-forces
- Motor shock
- Chassis vibration



MAJOR PROBLEM FOR NOW:

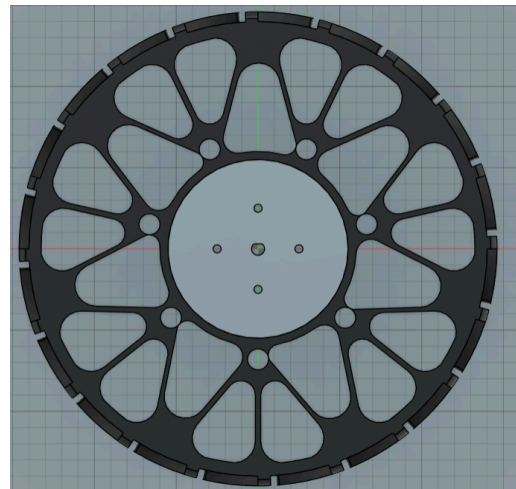
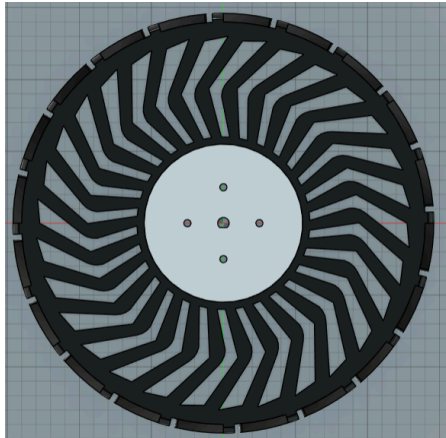
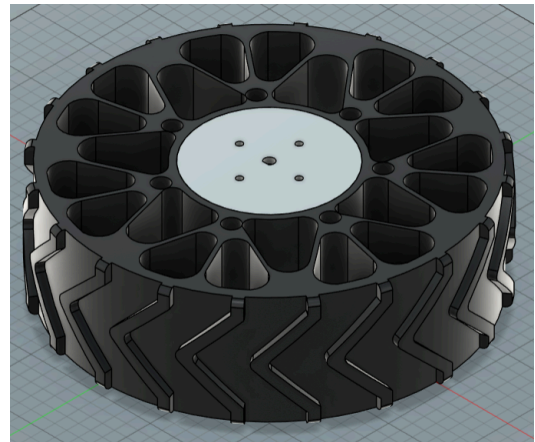
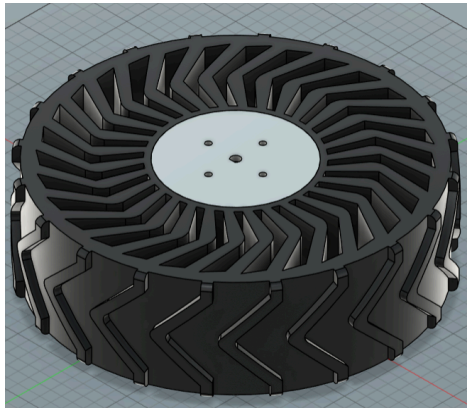
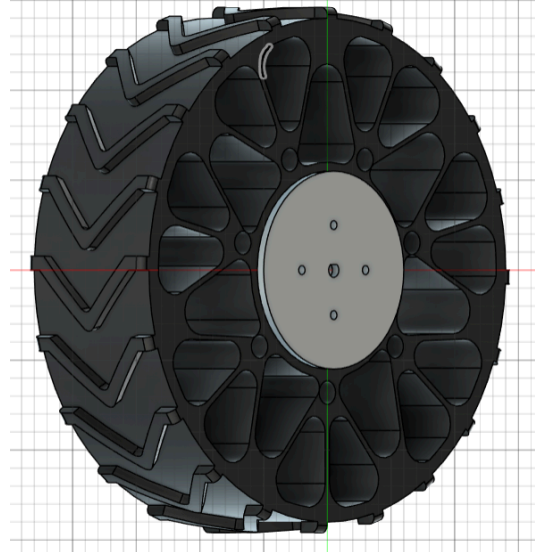
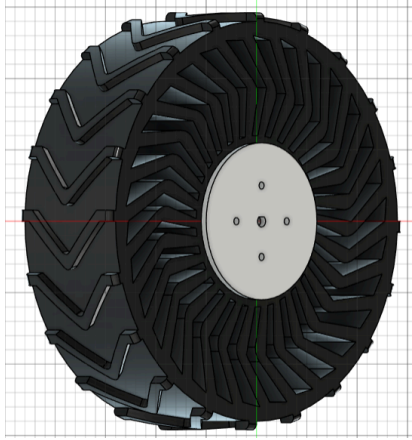
- Material of the HUB?? —> Based on cost, Robustness, feasibility, and how will we connect motor to the hub.....Coupler or some other method?

Solution 1 – Connect a thin metal (5mm) disc to the motor shaft and then join it to the hub of wheels using screws.

CAD updates on this task:

Iteration 1

Iteration 2 - changed tread design and Hub size increased + hollow hub



Iteration 3 - Spokes changed

STRESS ANALYSIS

